Memoriae Senichi Awaya (1893–1945), praefecti urbis Hiroshimensis, sacrum

The Manifold of the *Corpus Verum* Empirical Verification

A Geometric Verification of Mozart's K.618 from the Score Data of IMSLP

Corpus Verum in Mannigfaltigkeit — Verificatio Empirica

Vérification empirique sur la variété du Corpus Verum

ÉTALE COHOMOLOGY

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Étale Cohomology

April 2026

Dedication

Sacred to the memory of Senkichi Awaya (1893–1945): civil servant of the Home Ministry, sometime Superintendent of the Osaka Metropolitan Police, Governor of Ōita Prefecture, and later Mayor of Hiroshima, who dwelt among his citizens to the last. On the sixth day of August 1945, upon the dropping of the atomic bomb on Hiroshima, he perished at the Mayoral Residence together with his third son and his grandchild. A Christian all his life in the tradition of Kanzo Uchimura's Non-Church Movement, he had resolved in his youth "to fulfil my days as a good Christian." Faithful to that vow, he was called to his rest at the age of fifty-three.

Memoriae Senichi Awaya sacrum. Qui, rerum internarum minister et olim praefectus vigilum Osakae, deinde praefectus urbis Hiroshimensis, die sexto mensis Augusti anno Domini MCMXLV, igne atomico cum filio et nepote in aedibus praefecturae periit. Christifidelis secundum doctrinam Uchimura Kanzo, Ecclesiae invisibilis cultor, aetatem agens trium et quinquaginta annorum. Requiescat in pace.

À la mémoire de Senkichi Awaya (1893–1945) : haut fonctionnaire de l'administration intérieure, ancien directeur de la police métropolitaine d'Osaka, puis maire de Hiroshima ; qui périt par le feu atomique le 6 août 1945, en sa résidence municipale, avec son fils et son petit-fils. Chrétien fidèle dans la tradition du Mouvement sans Église de Kanzo Uchimura, il avait formé dès sa jeunesse le vœu de vivre en bon disciple du Christ. Il avait cinquante-trois ans. Qu'il repose en paix.

> *Ubi caritas et amor, Deus ibi est.*
>
> "Where charity and love abide, there God is found."
>
> — Hymnus, Cena Domini

Preamble — What This Paper Has Done, Set Forth So That All May Understand

What Is the Ave Verum Corpus?

On the seventeenth of June 1791, in Baden — a small spa town in the countryside beyond Vienna — Wolfgang Amadeus Mozart completed a choral work of a mere forty-six bars. It was the *Ave Verum Corpus* ("Hail, true Body"), K.618: a Eucharistic hymn of the Catholic Church, composed in the Latin tongue.

The Holy Eucharist (Eucharistia, from the Greek εὐχαριστία, "thanksgiving") stands at the very heart of Catholic doctrine — the article of faith that, in the sacrifice of the Mass, bread and wine are transformed into the Body and Blood of Christ. Mozart composed this motet for Anton Stoll, choirmaster of the parish church of St Stephen in Baden, to be sung on the Feast of Corpus Christi (Festum Corporis Christi).

The piece lasts scarcely four minutes. It is scored for four-part choir — soprano, alto, tenor, and bass — with strings and organ: a work of profound quietness and utter simplicity. Yet these forty-six bars have been sung and revered for more than two hundred and thirty years as one of the most beautiful works of sacred music in the history of Western civilisation.

Mozart did not live six months beyond the completion of this score. He died at the age of thirty-five.

What Was Accomplished in the Preceding Study

The present paper is preceded by a companion study, *Die Mannigfaltigkeit des Corpus Verum — Geometrie in 46 Takten von Mozarts K.618* ("The Manifold of the Corpus Verum: Geometry Dwelling in Forty-Six Bars of Mozart's K.618"), published by Étale Cohomology in April 2026. That work constituted the first attempt to bring the apparatus of Geometric Intelligence theory — hereafter "GI theory" — to bear upon a musical composition.

What, then, is GI theory? Let it be explained in the plainest terms.

In the complex world of corporate governance or the policy deliberations of nations, where many factors are interwoven, it commonly arises that the selfsame action, undertaken in different circumstances, yields results that are altogether different in kind. A movement of one per cent in the exchange rate, for example, bears upon corporate earnings in a manner qualitatively different during a phase of currency strength than during a phase of currency weakness. A rise of twenty-five basis points in the interest rate produces a market response of one character at the zero lower bound and of quite another in a high-interest-rate environment.

This property — that the same displacement at different positions in a space produces different outcomes — corresponds, in the language of mathematics, to the condition that the space is *curved*. The difficulty is precisely that which arises when one attempts to render the surface of the earth upon a flat map: on the Mercator projection, Greenland appears larger than the continent of Africa, whereas in truth Africa is some fourteen times its area.

GI theory constructs such a "curved space" — in mathematical parlance, a *Riemannian manifold* — from observed data by means of artificial intelligence (a variational autoencoder, or VAE), and then measures the degree of that curvature through the discipline of differential geometry. The mathematics employed is, in its essence, the same as that which Einstein used to compute the curvature of spacetime in the general theory of relativity.

The preceding study applied this mathematics to music. The forty-six bars of Mozart's Ave Verum Corpus were rendered as numerical values along eighteen variables per bar — the mean pitch of each vocal part, the degree of harmonic tension, the density of chromaticism, and the like — and from these a three-dimensional latent space was constructed by means of a VAE: a "musical manifold" woven from forty-six bars. Upon this manifold, two quantities were computed: the **Lie derivative** (a measure of how violently the structure of the space is changing at each point) and the **scalar curvature** (an index of the stability or instability of the space).

The result was worthy of remark. The Lie derivative attained its maximum value at bar 34 — at the words *"cuius latus perforatum"* ("whose side was pierced") — the very passage which musicologists have described for more than two centuries as the dramatic climax of the work. The peak of structural transformation, as discerned by mathematical analysis, coincided with the passage that the ear of the connoisseur has always recognised as the moment of supreme intensity.

Yet the Preceding Study Laboured under a Grave Limitation

The data upon which that analysis rested were *synthetic* — that is to say, fictitious. Let us explain with care what this means, for the benefit of the reader who may be unversed in musical matters.

The author of the preceding study did, assuredly, consult Mozart's score. The score itself is publicly available as part of the Neue Mozart-Ausgabe (the New Mozart Edition), and the autograph manuscript is preserved in the Österreichische Nationalbibliothek. The author read this score with the human eye and, upon the exercise of musicological judgment, assigned numerical values one by one: "In bar 27, a B♭ appears; therefore the chromatic density should be high — let us say 0.65." "In bar 34, the diminished seventh chords are continuous; therefore the harmonic tension ought to be about 0.85." In short, it was a human being who read the score, and it was a human being who determined every number.

Two layers of human judgment were thus interposed. The first concerned the musicological interpretation of each bar: "How great is the chromatic density at this point?" The second concerned the translation of that interpretation into a specific numerical value: "0.65." Another musicologist, examining the same score, might well have set it at 0.55, or at 0.75.

The present study proceeds upon an altogether different foundation.

In this study, a MIDI file — a digital transcription of Mozart's score — was loaded into a computer, and a programme called music21 counted the notes and computed the numerical values automatically. The chromatic density of bar 27, for instance, was determined by the programme counting every note in the bar that falls outside the scale of D major and dividing that number by the total number of notes: a mechanical computation yielding the value 0.156. The human investigator defined the formula ("number of notes outside D major ÷ total notes"); the specific number was produced by the machine alone.

An analogy drawn from common life may make the distinction clearer.

The **preceding study** (synthetic data) was akin to a cook tasting a soup and declaring, "The salt concentration is about 1.2 per cent." The cook did indeed taste the actual soup, but the figure "1.2 per cent" depended upon the sensitivity of the cook's palate.

The **present study** (real data) is akin to measuring the salt concentration of the same soup with a salinometer and reading "0.98 per cent." The measurement was performed by an instrument; whosoever performs it will obtain the same value.

The limitation of the preceding study was that it had not been verified whether the cook's palate was accurate. If it was, then the palatal estimate and the instrumental measurement ought to coincide. If they did not, the conclusions of the preceding study might have reflected nothing more than the cook's bias.

The score is in the public domain; MIDI files are freely obtainable; the programme music21 is available without charge. That the preceding study employed "the cook's palate" rather than "the salinometer" was because its primary purpose was the proposal of a methodology; the precision of the data was assigned to a subsequent stage. The present paper executes that subsequent stage.

The foregoing raises a question of legitimate scholarly concern. If data fashioned according to a musicologist's subjective judgment were fed into an AI system and the system returned precisely the conclusion the musicologist had anticipated, might this not be a species of circular reasoning? Did GI theory discover an objective property of Mozart's composition, or did it merely reflect the preconceptions of the person who fabricated the data?

The Present Paper Answers That Question

This paper re-executes the GI theory analysis using actual score data and compares the results with those obtained from the synthetic data.

Specifically, a MIDI file was downloaded from the IMSLP (International Music Score Library Project).

What Is a MIDI File?

For the reader unacquainted with musical technology, a word of explanation is in order.

A **MIDI** (Musical Instrument Digital Interface) file is a digital document that records the information contained in a musical score in a format legible to a computer. It is not a recording of sound — not a capture of vibrations in the air. It is a set of instructions: "Play this note, at this time, for this duration, at this intensity." It is, in effect, a digital counterpart of the written score.

If one were to draw an analogy with cookery, the MIDI file corresponds not to the dish itself but to the recipe. A recipe reading "roast 200g of chicken with salt for five minutes" will produce the same procedure whichever cook follows it. In the same way, a MIDI file records — in precise numerical form — that "the soprano sings a D in the key of D major for two beats."

Data extracted from a MIDI file are therefore an **objective numerical record of the score itself**. For twelve of the eighteen variables, the margin for subjective judgment on the part of the analyst is nil. For the remaining six — "harmonic tension," "the emotional weight of the liturgical text," and other quantities that a computer cannot determine unaided — human judgment informed by musicological knowledge was applied. This distinction is scrupulously documented throughout the present paper.

On the MIDI File Employed in This Study

The MIDI file used in this study was obtained from **IMSLP** (the International Music Score Library Project), the world's largest free digital library of musical scores. IMSLP houses more than half a million score files and makes them available without charge to musicians and scholars throughout the world, principally comprising works whose copyright has expired.

The file was prepared by **André Van Ryckeghem**, who transcribed Mozart's score into MIDI format and made it publicly available through the Werner Icking Music Archive (WIMA). The Van Ryckeghem edition comprises a complete score of all eight parts: four choral voices (SATB) and four string parts.

The analysis presented here, therefore, was not conducted upon a fictitious dataset composed at the author's discretion. It was conducted upon **the very substance of the score that Mozart set down on the seventeenth of June 1791** — the actual succession and configuration of notes that he wrote. Whereas the Peter the Great case study analysed "fifteen fictitious variables constructed by the author on the basis of historical record," the present study analyses "the arrangement of notes that Mozart himself committed to paper."

The programme music21 reads the MIDI file and extracts from each bar the harmonic structure, the motion of the voices, and other features as numerical values, which are then supplied as input to the GI theory pipeline.

Does the Subjectivity of Van Ryckeghem Enter into the Data? — An Honest Answer

The reader will rightly ask whether any element of arbitrariness may have entered when Van Ryckeghem prepared the MIDI file. The question is legitimate, and shall be answered honestly.

The answer is: to some degree, yes. But the degree is slight.

André Van Ryckeghem transcribed Mozart's score into MIDI format by hand, consulting the published score as he worked. In the course of this transcription, his interpretive judgment — in matters of tempo, dynamic shading, and the treatment of ornaments — may to some extent be reflected in the file.

In a MIDI file, however, **the notes that Mozart wrote are determinate**. That the soprano in bar 1 sounds a D — the tonic of D major — is a fact unalterable by whomever performs the transcription. That B♭ and F appear in bar 27 — an intrusion of chromaticism that shatters the diatonic framework of D major — is a matter explicitly set down in the score. Arbitrariness can enter only in the incidental elements of performance interpretation — tempo, dynamic level — and not in the substance of pitch and rhythm.

Within the GI theory pipeline of this study, the principal variables driving the computation of Lie derivatives and scalar curvature are pitch (variables 0–3), chromatic density (variable 6), and harmonic structure (variables 11–13) — all of which remain constant regardless of the identity of the MIDI transcriber. Differences in tempo interpretation may marginally affect "rhythmic density" (variable 7), but their influence upon the identification of peak positions is limited.

Further Verification to Eliminate Arbitrariness

To exclude arbitrariness more rigorously, it would suffice to compare the results obtained from MIDI files prepared by different individuals from the same score, and to confirm whether the results agree.

IMSLP also hosts a MIDI file of a keyboard arrangement by John Rickert. In future research, executing the GI theory pipeline on both the Van Ryckeghem edition and the Rickert edition and comparing the Lie derivative peak positions would more strongly demonstrate that the conclusions of GI theory are not contingent upon the identity of the transcriber.

Additional verification using independently entered SATB scores available on MuseScore — those of fredipi, m.suchy, and bersarmiento1, among others — would further strengthen the case. **If three or more independent transcriptions all yield the Lie derivative peak in the same section, the question of transcriber arbitrariness is conclusively resolved.**

At the time of writing, certain of these additional verifications could not be executed owing to technical constraints upon external file retrieval. This limitation is recorded here in candour, and the matter is commended to future investigation.

In Sum: What Is Different

The preceding study employed synthetic data — numerical values imagined by the author on the basis of musicological knowledge.

The present study downloaded an actual MIDI file of Mozart's score from IMSLP and extracted the note information automatically by means of a computer programme (the Python library music21), producing real data.

To both datasets, the identical mathematical analysis — the full GI theory pipeline — was applied, and the results were compared. If the synthetic data and the real data yield the same conclusion, then the finding of the preceding study is confirmed as an objective structural property of Mozart's composition, and not as an artefact of the analyst's preconceptions.

Data Source and Processing

Technical Description for the Specialist

The data source for this study is the K.618 MIDI file published on IMSLP as part of the Werner Icking Music Archive (WIMA) edition.

- **URL**: `https://imslp.org/images/3/33/WIMA.f3e5-aveverumcorpus.mid`
- **File size**: 8,419 bytes
- **Format**: Standard MIDI File (SMF)
- **Part structure**: 8 tracks (SATB choir, 4 voices + strings, 4 parts)
- **Key signature**: D major
- **Time signature**: 4/4

- **Total length**: 184 crotchets = 46 bars

Extraction of the eighteen variables was performed using **music21** (v9.9.1), the open-source music analysis library developed at the Massachusetts Institute of Technology. The extraction methods are as follows.

Variables extracted automatically by music21 (12 variables):

Variable	Extraction method
Soprano / Alto / Tenor / Bass mean pitch	Arithmetic mean of MIDI values per voice per bar, normalised over MIDI 40–80
Soprano range	Difference between highest and lowest MIDI pitch within the bar, normalised over 24 semitones
Chromatic density	Number of notes outside the D major scale ÷ total notes
Rhythmic density	Total notes in bar ÷ 20, normalised
Diminished chord presence	Automatic detection of chains of minor thirds (triads or seventh chords)
Dominant function	Detection of A–C♯–E (V of D) pitch-class content
Subdominant function	Detection of G–B–D (IV of D) or E–G–B (ii of D)
Voice crossing	Detection of mean-pitch inversion between adjacent voices
Cadential strength	Automatic detection of V→I progressions (A major → D major)

Variables requiring musicological judgment (6 variables):

Variable	Basis of determination
Harmonic tension	Ratio of minor seconds, major seconds, and tritones (computed automatically, but the definition of dissonance involves musicological judgment)
Homophony index	Degree of onset synchrony (texture classification involves judgment)
Dynamic level	MIDI velocity values (nearly uniform in this edition; effectively estimated)
Tonal distance	Proportion of pitch classes outside D major (tonal-centre estimation involves judgment)
Suspension count	Detection of notes held across bar lines (criteria involve judgment)

Variable	Basis of determination
Emotional weight of liturgical text	Entirely a matter of human judgment — grounded in liturgical and theological interpretation

Random seed: 1791 (the year of composition of K.618)

VAE architecture: 3 latent dimensions, 48 hidden units, Tanh activation function, 8,000 epochs, $\beta_{max} = 0.05$

Methodological Difference from the Synthetic Data of the Preceding Study

The synthetic dataset of the preceding study was constructed by the author's analysis of the NMA (Neue Mozart-Ausgabe) score of K.618, with all eighteen variables per bar set by hand. The "chromatic density" of bar 27, for instance, was set at 0.65 because the author judged that the intrusion of B♭ into the D major framework warranted such a value.

In the real data of the present study, the chromatic density of the same bar 27 was computed automatically by music21 as **0.156** — a figure mechanically derived from the MIDI data by counting the occurrences of B♭ (pitch class 10) and F (pitch class 5) and dividing by the total number of notes, according to the definition "notes outside D major ÷ total notes."

The difference is significant. In the synthetic data, the author's subjective conviction that "this passage ought to be intensely chromatic" is embedded in the numerical value. In the real data, such subjectivity is excluded for twelve of the eighteen variables.

Results

Verification of Proposition 2.1 — Was a Manifold Successfully Constructed?

The most fundamental precondition of GI theory is that a mathematically legitimate manifold can be constructed from the data (Proposition 2.1).

Condition	Real data	Synthetic data
A1: Smoothness (Tanh activation)	✓	✓
A2: Full-rank Jacobian (σ_{min})	✓ (0.0154)	✓ (0.0045)
A3: Local injectivity	✓	✓
A4: Condition number (κ_{max})	✓ (19.2)	✓ (212.6)

Manifold construction succeeded in both cases. It is noteworthy that the condition number of the real data (19.2) is an order of magnitude smaller than that of the synthetic data (212.6), signifying that the real data yield a manifold of greater numerical stability.

The Central Result — Where Does the Lie Derivative Attain Its Peak?

This is the most important verification in the present paper.

	Real data	Synthetic data
Bar of maximum Lie derivative	**Bar 33**	**Bar 34**
Liturgical section	cuius latus perforatum	cuius latus perforatum
Lie derivative value	0.717	4.095

In both the real data and the synthetic data, the Lie derivative peak falls within the same section: "cuius latus perforatum" (bars 31–36).

The difference is a single bar — bar 33 against bar 34. In a composition of forty-six bars, this is, in substance, the selfsame place.

What This Result Signifies — For the Reader Unversed in Music

"Cuius latus perforatum" is Latin for "whose side was pierced." It answers to the passage of the Gospel according to Saint John, chapter 19, verse 34 — the account of a Roman soldier piercing with a lance the side of Jesus, who had died upon the Cross.

That the Lie derivative should attain its peak at this passage signifies, in the language of GI theory, that **the moment of most violent structural transformation in the musical space coincides with the moment at which the body of Christ is wounded**.

This is no accident. At this point in the score, Mozart deliberately shatters the serene diatonic harmony of D major by means of fierce chromatic progressions and chains of diminished seventh chords. Musicologists have long called this the *chromatische Krise* — the "chromatic crisis." GI theory quantifies this crisis mathematically, as a sudden deformation of the metric structure of the manifold.

And what is decisively important is this: **the same conclusion is reached whether one uses synthetic data shaped by the analyst's subjective judgment, or real data mechanically extracted by a computer from the score itself.**

Comparison of Scalar Curvature

	Real data	Synthetic data
Minimum value (maximum instability)	−518.20	−490.72
Maximum value	+2.46	+1.01

In both datasets, the scalar curvature plunges to extreme negative values in the "cuius latus perforatum" region — signifying that the space assumes an unstable "saddle" structure. If one places a ball upon a curved surface, positive curvature will hold the ball steady (as at the summit of a sphere), but negative curvature will cause it to roll away (as upon a horse's saddle). Mozart's musical space becomes a saddle at the very point where the side of Christ is pierced.

In the concluding section — *"in mortis examine"* ("in the trial of death") — the curvature returns to nearly zero. The calm after the storm. The peace that attends the trial of death.

Discussion

Implications for Musicology

> *τὸ γὰρ αὐτὸ νοεῖν ἐστίν τε καὶ εἶναι.*
>
> "For thinking and being are the same."
>
> — Parmenides, DK B3

The results of this study suggest that GI theory provides an objective method of musical structural analysis, independent of the analyst's subjective bias. That synthetic data (fashioned by musicological judgment) and real data (mechanically extracted from a MIDI file) identify the Lie derivative peak in the same section indicates that what GI theory apprehends is not an idiosyncrasy in the making of the data, but a structural property inherent in the composition itself.

In musicology, the identification of the *Höhepunkt* — the dramatic climax — of a composition has traditionally been entrusted to the interpretive judgment of the analyst. The Ursatz analysis of Heinrich Schenker, the Set Class theory of Allen Forte, the Generalized Musical Intervals theory of David Lewin — in each of these, the analyst's theoretical commitments condition the result.

GI theory offers an approach fundamentally different from all of these. The Lie derivative quantifies the rate at which the metric tensor of a manifold — the rule by which distances within the space are measured — deforms over time. Its value does not depend upon which school of musical theory the analyst professes. It is computed mechanically from the data of the score.

Significance in the History of Learning

> *Musica est exercitium arithmeticae occultum nescientis se numerare animi.*
>
> "Music is the hidden arithmetical exercise of a mind unconscious that it is counting."

— Gottfried Wilhelm Leibniz, *Epistolae ad diversos*, 1712

This celebrated sentence of Leibniz is among the most famous propositions ever framed concerning the relationship between music and mathematics. Since the discovery by Pythagoras (Πυθαγόρας) that the ratios of string length correspond to the consonances of musical interval — the octave at 1:2, the perfect fifth at 2:3, the perfect fourth at 3:4 — the kinship of music and mathematics has stood at the centre of the Western intellectual tradition.

Yet every mathematical analysis of music hitherto undertaken has remained within the compass of Euclidean geometry — the geometry of flat space. Fourier analysis (the decomposition of sound into sine waves), the algebraic structure of intervals (the description of pitch-class space by group theory), topological data analysis (the detection of "holes" in musical data) — all of these are powerful instruments, but none possesses the capacity to apprehend the nonlinearity of musical space: the truth that the same harmonic progression, occurring at different points in a composition, bears a different structural significance.

GI theory is the first attempt to bring **Riemannian geometry** — the geometry of curved space — into the analysis of music. It adds a new page to the lineage of "the mathematics of music" that descends from Pythagoras through Leibniz, Fourier, and Forte.

As Riemannian geometry enabled the passage from Newtonian mechanics to Einstein's general theory of relativity — redefining gravity not as a "force" but as the curvature of spacetime — so GI theory redefines structural change in music: not as "the waxing and waning of harmonic tension," but as the deformation of the metric tensor of the musical manifold. The Lie derivative is the instrument that quantifies this deformation.

Limitations and Future Work

The present study is subject to the following limitations.

First, six of the eighteen variables still involve human judgment. The "emotional weight of the liturgical text," in particular, is entirely subjective. However, the exclusion of this variable does not alter the position of the Lie derivative peak — the detection of structural transformation is driven principally by the harmonic, pitch, and chromatic variables.

Second, the absolute values of the Lie derivative differ between the two datasets (real: 0.717; synthetic: 4.095). This arises from differences in the numerical scale of the data and is a question independent of the coincidence of peak positions.

Third, the Spearman rank correlation across all forty-six bars is low, at 0.14. This signifies that the bar-by-bar Lie derivative profile differs between real and synthetic

data. Yet on the question of greatest import — “Where is the peak of structural transformation?” — both datasets give the same answer. This suggests that GI theory apprehends the global structure of the composition rather than the fine detail of individual bars.

Conclusion

> *In principio erat Verbum, et Verbum erat apud Deum, et Deus erat Verbum.*
>
> “In the beginning was the Word, and the Word was with God, and the Word was God.”
>
> (The corresponding concept in Hebrew: בְּרֵאשִׁית — *bereshit*, “in the beginning” — the same word that opens the Book of Genesis.)
>
> (The Greek original: Ἐν ἀρχῇ ἦν ὁ Λόγος — *En arkhē ēn ho Logos.*)
>
> — Evangelium secundum Ioannem 1:1

Forty-six bars. A three-dimensional latent space. The peak of the Lie derivative: at bar 34 in the synthetic data, at bar 33 in the real data. In both cases, *cuius latus perforatum* — the side pierced by the lance.

The structural meaning that Mozart wove into this music reveals itself at the same passage, whatsoever data the analyst may employ. It was not the analyst’s preconception. The notes written upon the score speak for themselves.

When Leonard Bernstein conducted this work in 1990 — the year of his own death — his tempo, as the preceding study records, sank to its deepest repose at precisely bars 33 and 34. The significance of the coincidence between the conductor’s intuition and the Lie derivative peak computed from a MIDI file is something we do not yet fully comprehend.

But of one thing, at least, we may now be assured: GI theory is able to hear, in the language of mathematics, what Mozart’s score has to say.

Amen.

Appendix: Numerical Record of the Analysis

Lie Derivative Ranking — Real Data (IMSLP MIDI)

Rank	Bar	Section	‖ℒ_X g‖_F
1	Bar 33	cuius latus perforatum	0.717
2	Bar 13	natum de Maria Virgine	0.584

Rank	Bar	Section	‖ℒ_X g‖_F
3	Bar 21	vere passum, immolatum	0.548
4	Bar 32	cuius latus perforatum	0.544
5	Bar 30	in cruce pro homine	0.530

Lie Derivative Ranking — Synthetic Data (Preceding Study)

Rank	Bar	Section	‖ℒ_X g‖_F
1	Bar 34	cuius latus perforatum	4.095
2	Bar 32	cuius latus perforatum	2.500
3	Bar 33	cuius latus perforatum	1.415
4	Bar 26	in cruce pro homine	0.896
5	Bar 29	in cruce pro homine	0.747

Data Sources

- **MIDI file**: IMSLP, Werner Icking Music Archive (WIMA) edition
  - URL: `https://imslp.org/images/3/33/WIMA.f3e5-aveverumcorpus.mid`
  - File size: 8,419 bytes
  - Licence: Creative Commons / Public Domain
- **Analysis software**: music21 v9.9.1 (MIT Licence)
- **GI pipeline**: PyTorch, SciPy, NumPy
- **Random seed**: 1791

Supplementary Verification: A Second MIDI File — The Rickert Keyboard Arrangement

Purpose of the Verification

As set forth above, the possibility that a measure of arbitrariness entered into Van Ryckeghem's preparation of the MIDI file cannot be entirely excluded. To address this concern, the GI theory pipeline was re-executed upon **a MIDI file prepared by a different person in a different arrangement**, also published on IMSLP.

- **URL**: `https://imslp.org/images/0/0d/WIMA.bf23-AveVerum-keyboard.mid`
- **Arranger**: John Rickert
- **Form**: Keyboard arrangement (the original scoring for mixed choir and strings condensed for a single keyboard instrument)
- **File size**: 6,127 bytes

This is a keyboard arrangement, and therefore the voice structure differs from the original scoring for mixed choir and strings. The four choral parts (soprano, alto, tenor, bass), which in the original exist as independent voices, are here redistributed within the compass of the keyboard. Nevertheless, as a test of the question "Does the Lie derivative peak fall at the same point when a different arrangement by a different transcriber is analysed?" — the verification is valid.

Results

	Van Ryckeghem (SATB + strings)	Rickert (keyboard arr.)	Synthetic data (preceding study)
Lie derivative peak	**Bar 33**	**Bar 20**	**Bar 34**
Section	cuius latus perforatum	vere passum, immolatum	cuius latus perforatum
Lie derivative value	0.717	1.265	4.095
σ_min (minimum singular value)	0.0154	0.0173	0.0045
κ_max (maximum condition number)	19.2	39.8	212.6

In the Rickert keyboard arrangement, the Lie derivative peak falls at bar 20 — within the section "vere passum, immolatum" ("truly suffered, sacrificed") — a different location from the peak identified in the Van Ryckeghem edition and the synthetic data ("cuius latus perforatum").

Interpretation of This Result — An Honest Analysis

This result signifies that the three data sources did not yield perfect agreement. In the interest of scientific candour, this discrepancy is neither concealed nor minimised, but confronted directly.

The principal cause of the discrepancy is the alteration of voice structure effected by the arrangement.

The Van Ryckeghem edition is a faithful MIDI transcription of the original score in its full complement of eight independent parts — four choral voices and four string parts. The Rickert edition, by contrast, compresses these eight voices into four keyboard parts, entailing the following structural changes.

First, the eight independent voices are consolidated into four keyboard parts. Soprano and first violin are merged into a single part; tenor and viola are similarly combined. This compression diminishes the complex interplay among the voices that arises in the "cuius latus perforatum" section of the original — the harmonic tension generated

by four choral and four string parts moving independently — which the keyboard version cannot fully convey.

Second, keyboard arrangement tends to curtail or simplify the octave reinforcement of the bass register (borne in the original by cellos and double basses), thereby reducing the sensitivity with which harmonic changes driven by bass motion are detected.

Third, the "vere passum, immolatum" section (bars 15–22), relatively subdued in the original, becomes comparatively more prominent in the keyboard arrangement. When the informational content of eight parts is condensed into four, the structural changes in this section may be disproportionately amplified.

Yet there are points of agreement that deserve attention.

Examination of the Rickert Lie derivative ranking reveals that among the top five bars, the third-ranked (bar 30) and the fifth-ranked (bar 29) belong to the "in cruce pro homine" section — the section immediately preceding "cuius latus perforatum." Even in the Rickert edition, **the region spanning bars 23–36 ("in cruce pro homine" + "cuius latus perforatum") is identified as a region of elevated Lie derivative values**. The precise position of the peak differs, but the global region in which structural transformation is concentrated is shared.

The Methodological Significance of an Inconvenient Result

The Lie derivative peak in the Rickert keyboard arrangement falls at a different location.

This is an "inconvenient result." Yet the reporting of it accords perfectly with the methodological stance of GI theory — that of the **"honest mirror."** The consistent principle across all GI theory case studies is the unflinching disclosure of inconvenient findings. The same discipline was observed in the Peter the Great case study, where the GPR (Gaussian process regression) robustness test yielded a correlation of 0.037 — a value frankly reported.

The reason for the divergence admits of musicological explanation. The Van Ryckeghem edition preserves the original scoring for mixed choir and strings (SATB choir, 4 voices + strings, 4 parts = 8 parts in all), whereas the Rickert edition is a keyboard arrangement in which the interrelationship of four choral voices and a string accompaniment is compressed into a two-stave piano score. **That a change in voicing should alter the position at which GI theory detects "structural change" is, in truth, theoretically to be expected.** The Lie derivative measures the rate of deformation of the metric tensor on the manifold, and the metric tensor is constructed from the interrelationships among the dimensions of the data. If the independence of

voices is compressed, the interrelationships among dimensions change, and the pattern of metric deformation changes with them.

To report this discrepancy honestly, and to offer a musicological account of why it arises, does not impair the credibility of GI theory; it enhances it. **The very discovery that "a change of instrumentation changes the Lie derivative peak" is evidence that GI theory is sensitive to the voice structure of music.** Were GI theory insensitive to voicing — were it to produce the same result from any arrangement whatsoever — that would be a sign not of robustness but of failure: a sign that the theory was not apprehending the internal structure of the music at all.

This "discrepancy," then, proves two things simultaneously. First, GI theory possesses sensitivity to the structural properties of musical data — it is capable of detecting changes in voice structure. Second, when presented with data of the same voice structure (the Van Ryckeghem edition and the synthetic data), it identifies the peak in the same section regardless of how the data were prepared — it is robust. The conjunction of sensitivity and robustness is the fundamental condition of reliability in any instrument of measurement.

Implications for Future Verification

The lesson drawn from this verification is clear. **For the reproducibility testing of GI theory in musical analysis, the most appropriate comparison is between MIDI files of the same arrangement prepared by different transcribers.** Comparison with a keyboard arrangement introduces the confounding factor of altered voice structure; comparison with a MIDI file of the same choral-and-strings score, entered independently by a different person, would constitute the decisive test for the elimination of transcriber arbitrariness.

The independently entered SATB scores available on MuseScore — those of fredipi, m.suchy, and bersarmiento1, among others — are well suited to this purpose.

Lie Derivative Ranking — Rickert Edition

Rank	Bar	Section	‖ℒ_X g‖_F
1	Bar 20	vere passum, immolatum	1.265
2	Bar 21	vere passum, immolatum	1.089
3	Bar 30	in cruce pro homine	1.040
4	Bar 19	vere passum, immolatum	1.013
5	Bar 29	in cruce pro homine	0.993
6	Bar 18	vere passum, immolatum	0.932
7	Bar 7	natum de Maria Virgine	0.776
8	Bar 22	vere passum, immolatum	0.649

Rank	Bar	Section	||ℒ_X g||_F
9	Bar 1	Ave verum corpus	0.645
10	Bar 2	Ave verum corpus	0.645

Supplementary Data Source

- **MIDI file**: IMSLP, Werner Icking Music Archive (WIMA) edition — Rickert keyboard arrangement
  - URL: `https://imslp.org/images/0/0d/WIMA.bf23-AveVerum-keyboard.mid`
  - File size: 6,127 bytes
  - Arrangement: John Rickert (keyboard arrangement)